Project 1: Array Parameter Estimation

EE4715: Array Processing
Delft University of Technology

Burcu Erarslanoglu (5948355)   Joris van de Weg (5174821)

1 Introduction

In this paper, crucial aspects of array processing, focusing on the estimation of directions and frequencies of sinusoidal signals entering a uniform linear array (ULA) and channel equalization are explored. For the first part, the aim is to estimate directions and frequencies based on a data model which simulates received signals using a ULA with multiple antennas. Then, three algorithms are implemented for the estimation of directions and frequencies and compared in terms of their performance. The performance of these algorithms is compared while considering different signal-to-noise ratios (SNR). Then, zero-forcing beamformers based on these direction and frequency estimates are evaluated to comment on their effectiveness in interference suppression. For the second part, a channel equalization problem is tackled by first modelling and reconstructing transmitted data symbols over a QPSK-modulated channel. This involves constructing a data matrix from sampled received signals and applying both zero-forcing and Wiener equalizers to analyze their performance. Overall, this paper illustrates the application of theoretical concepts to practical scenarios and enhances our understanding of advanced signal processing techniques used in array processing systems.

2 Estimation of Directions and Frequencies

Accurately estimating the directions-of-arrival (DOAs) and frequencies of incoming signals is crucial for applications regarding radar, wireless communication systems, and many others. The incoming signals usually originate from a combination of different sources which are measured by the system of antennas. This paper will showcase and evaluate the estimation of the d complex sinusoidal signals and their directions and frequencies using a ULA layout. A ULA, as the name already indicates, is a collection of identical antennas that are equally separated in a straight line. This spacing between the antennas is denoted as Δ. Using this spatial configuration and the fact that the signals propagate as wavefronts, more information can be extracted from the data. The problem setup is illustrated below in Figure 1.

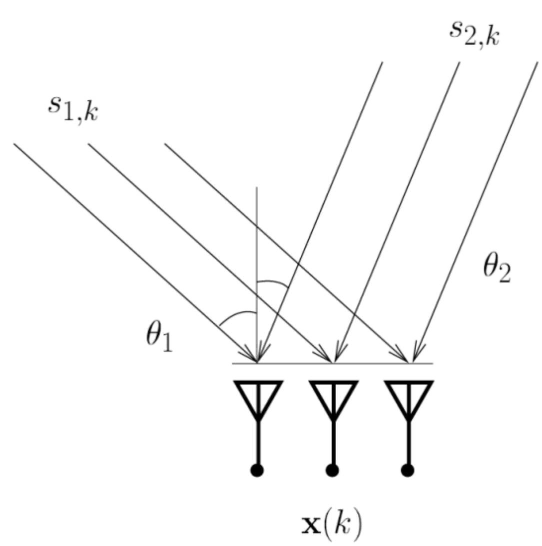

Figure 1: Illustration of the DOAs θ_1 and θ_2 of incoming signals $s_{1,k}$ and $s_{2,k}$, being measured by an array of antennas, represented by $\mathbf{x}(k)$.

In the problem setup given, M antennas, and N samples are considered. The antenna array of ULA is with a separation of $\Delta = 1/2$ wavelengths. Moreover, the parameters of interest are stored in vectors $\boldsymbol{\theta} = [\theta_1, \cdots, \theta_d]^T$ and $\mathbf{f} = [f_1, \cdots, f_d]^T$, containing the directions of the sources in degrees as $-90 \leq \theta_i < 90$ and the normalized frequencies of the sources as $0 \leq f_i < 1$, respectively. Finally, a temporally and spatially white, complex Gaussian noise is considered in the medium and with that a signal-to-noise (SNR) value per source is defined. To obtain the desired SNR per source, the noise is scaled. The i^{th} source is a complex sinusoid with normalized frequency f_i and it is given by:

$$s_{i,k} = e^{j2\pi f_i k}. \tag{1}$$

When looking more carefully at how these signals arrive at the antennas from Figure 1, a geometric relationship can be found between the angle of arrival and the phase shift caused by this angle. Figure 2 presents this relationship.

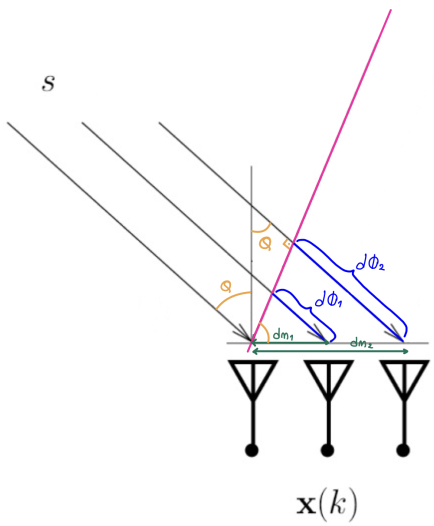

Figure 2: Illustration of the problem setup by showing the trigonometric relations.

As it can be seen in Figure 2, the incoming signal s arrives at the array with an angle θ in wavefront propagation. Carrying this θ shows us the trigonometric relation of $\sin(\theta) = d\phi_1/dm_1 = d\phi_2/dm_2$. When this relationship is extended to the scenario where multiple signals and antennas are present, the time delay for the m^{th} antenna in the array can be written relative to the first antenna as:

$$\tau_{i,m} = m \cdot \frac{\Delta\lambda \sin(\theta_i)}{c} \tag{2}$$

where the c is the speed of light. The phase shift ϕ_i corresponding to this time delay for a signal with frequency f where angular frequency is $\omega = 2\pi f$ is given by $\phi_i = \omega\tau_i = 2\pi f \tau_i$. Substituting $\tau_{i,m}$ and extending to the multi signal and antenna scenario, the phase shift for the m^{th} antenna in the array can be written relative to the first antenna as:

$$\phi_{i,m} = m \cdot 2\pi f \cdot \tau_i = m \cdot 2\pi f \cdot \frac{\Delta\lambda \sin(\theta_i)}{c} = m \cdot 2\pi \cdot \Delta \sin(\theta_i). \tag{3}$$

According to these, for the first antenna $m = 0$ and for the last antenna $m = M-1$. The derivations for the time delay in eq. (2) and phase shift in eq. (3) will be used in the next part of the paper where the received signal matrix $\mathbf{X}$ of size $M \times N$ is generated.

2.1 Signal Model

The signal received at the m^{th} antenna (after demodulation by ω_0) can be described with the following expression:

$$x_m(t) = a_m(\theta) \cdot s_0(t - T_m)e^{-j\omega_0 T_m} \tag{4}$$

This expression forms the basis for our signal model which is characterized by the received signal matrix $\mathbf{X}$. To generate this matrix of received signals for each microphone, the array response vector needs to be computed. The array response refers to the response of a particular antenna and describes the phase shifts experienced by a signal propagating from direction θ_i. The following expression for the array response is written for the i^{th} source by using the phase shift term eq. (3):

$$\mathbf{a}(\theta_i) = \begin{bmatrix} 1 \\ e^{j2\pi\Delta\sin(\theta_i)} \\ e^{j4\pi\Delta\sin(\theta_i)} \\ \vdots \\ e^{j2\pi(M-1)\Delta\sin(\theta_i)} \end{bmatrix}. \tag{5}$$

The response is arranged in rows to indicate which antenna received the response. There are d different sources which results in the array response matrix $\mathbf{A}$ of size $M \times d$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}(\theta_1) & \mathbf{a}(\theta_2) & \cdots & \mathbf{a}(\theta_d) \end{bmatrix}. \tag{6}$$

The incoming signals are described by the source signal matrix $\mathbf{S}$. It is constructed by stacking the source signals for all sources, with their corresponding frequencies, and all samples. The matrix $\mathbf{S}$ of size $d \times N$ is written as follows:

$$\mathbf{S} = \begin{bmatrix} \exp(j2\pi f_1 \cdot 0) & \exp(j2\pi f_1 \cdot 1) & \cdots & \exp(j2\pi f_1 \cdot (N-1)) \\ \exp(j2\pi f_2 \cdot 0) & \exp(j2\pi f_2 \cdot 1) & \cdots & \exp(j2\pi f_2 \cdot (N-1)) \\ \vdots & \vdots & \ddots & \vdots \\ \exp(j2\pi f_d \cdot 0) & \exp(j2\pi f_d \cdot 1) & \cdots & \exp(j2\pi f_d \cdot (N-1)) \end{bmatrix} \tag{7}$$

By using the matrix expressions given in eq. (6) and eq. (7), the received signal matrix $\mathbf{X}$ of size $M \times N$ for the entire layout with M antennas and N time samples is written as follows:

$$\mathbf{X} = \mathbf{AS} \tag{8}$$

Furthermore, each element $x_{m,n}$ of $\mathbf{X}$ can be written as:

$$x_{m,n} = \sum_{i=1}^{d} e^{j2\pi m\Delta\sin(\theta_i)} \cdot \exp(j2\pi f_i \cdot n) \tag{9}$$

Later, a temporally and spatially white, complex Gaussian noise matrix $\mathbf{N}$ can be added to the received signal matrix. The signal power is calculated using the given SNR and the noise variance is found by using a noise power of 1 divided by the found signal power. The noise matrix $\mathbf{N}$ is generated to have both real and imaginary parts and it is added directly to eq. (7) as $\mathbf{X} = \mathbf{AS} + \mathbf{N}$.

Singular value decomposition (SVD) may provide a lot of insight into the construction of this problem because with singular values, it is able to comment on the strength of signals, and the separation between the signal subspace from the noise subspace. Changes in the array configuration can be analyzed from singular values, as well. Firstly, the singular values of $\mathbf{X}$, for $d = 2$ sources, $M = 5$, $N = 20$, $\boldsymbol{\theta} = \begin{bmatrix} -20, 30 \end{bmatrix}^{\mathrm{T}}$, $\mathbf{f} = \begin{bmatrix} 0.1, 0.3 \end{bmatrix}^{\mathrm{T}}$, and SNR $= 20\,$dB are plotted in Figure 3 below.

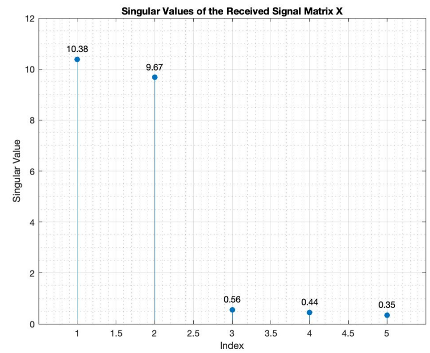

Figure 3: The singular values of the received signal matrix $\mathbf{X}$.

When Figure 3 is analyzed, it is seen that the first two singular values are significantly larger than the others. These two large singular values correspond to the two sources in the setup and indicate that there are two dominating signal components. The other three singular values represent the noise subspace. The fact that they are significantly smaller than the first two singular values suggests a clear separation between the signal and noise subspaces and shows that the array configuration with an SNR of 20 dB is effective in resolving the signals coming from the two sources.

To observe the effect of setup parameters on the singular values, the following scenarios were considered: doubling the number of samples to $N = 40$, doubling the number of antennas to $M = 10$, reducing the angle between the sources to $\boldsymbol{\theta} = \left[-20, -10\right]^{\mathrm{T}}$, and reducing the frequency difference to $\mathbf{f} = \left[0.1, 0.11\right]^{\mathrm{T}}$. The results can be observed in Figure 4 below.

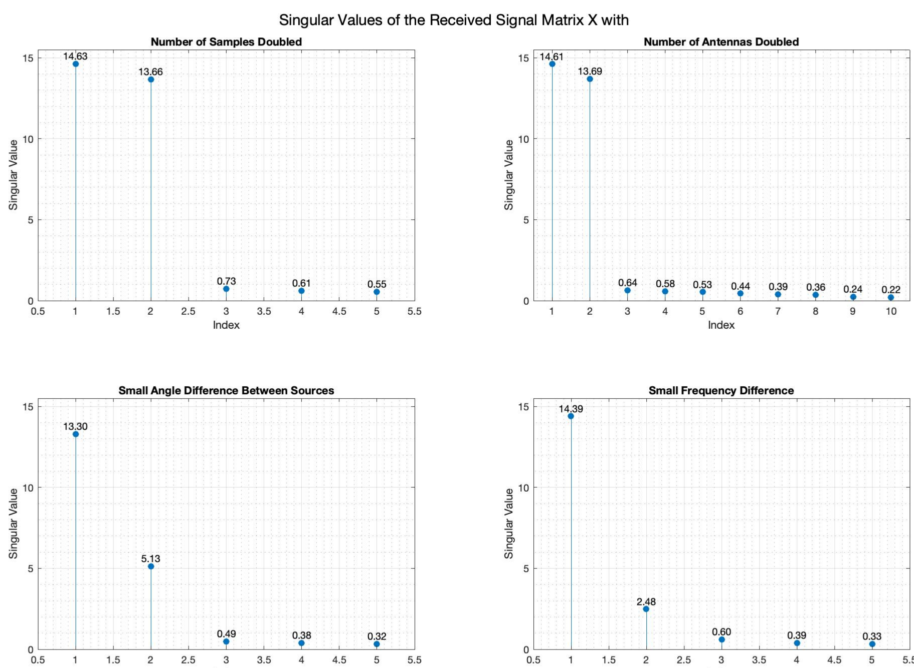

Figure 4: The singular values of the received signal matrix $\mathbf{X}$ for different setup parameters.

When Figure 4 is analyzed in comparison with Figure 3 it is seen that for the case where the number of samples N are doubled, the singular values became slightly larger due to the improved signal averaging. This makes the distinction between the signal and noise subspace more pronounced. For the case where the number of antennas M are doubled, it is first seen that there are 10 singular values instead of 5. This is because the size of the matrix $\mathbf{X}$. Again, the value of the first two singular values in Figure 4 are larger than the ones in Figure 3 because the spatial resolution of the ULA is improved when more antennas are added. For the bottom two plots where the effect of angle and frequency is analyzed poorer performance can be seen overall. For the case where the angle between the sources is reduced, the separation between the signal and noise subspace less pronounced because the gap between the second and third singular value is now less. Also, the second singular value is now a lot less than the first one because is is more challenging for the array to distinguish the sources in the presence of noise. Finally, analyzing the effect of reducing the frequency difference yields similar results. The difference between the sources is harder to distinguish based on frequency.

Overall, doubling the number of samples and antennas enhances resolution and signal-noise separation, while reducing the angle and frequency differences between sources diminishes resolution, making it harder to distinguish between sources.

2.2 Estimation of Directions

When estimating the DOAs, different algorithms with various strengths, such as the MVDR, MUSIC, and ESPRIT algorithms can be implemented. The focus of this part is the ESPRIT algorithm which estimates the DOAs by relying on a special array configuration of the data matrix $\mathbf{X}$. This allows the DOAs to be estimated algebraically by solving an eigenvalue problem. This property distinguishes the ESPRIT algorithm from MVDR or MUSIC because it does not require a search or calibration data.

For the ESPRIT algorithm, the structure of the eq. (5) can be exploited by using the shift-invariance property. When $\phi_i = e^{j2\pi\Delta\sin(\theta_i)}$, the following partition on $\mathbf{a}(\theta_i)$ can be made:

$$\boldsymbol{a}_x := \begin{bmatrix} 1 \\ \phi \\ \vdots \\ \phi^{M-2} \end{bmatrix}, \quad \boldsymbol{a}_y := \begin{bmatrix} \phi \\ \phi^2 \\ \vdots \\ \phi^{M-1} \end{bmatrix}, \quad \text{so that} \quad \boldsymbol{a}_y = \boldsymbol{a}_x \phi. \tag{10}$$

Because the same structure is also present in matrix $\mathbf{X}$ given in eq. (8), this matrix can be grouped so that the first and last $M-1$ antennas are separated. Which gives the two matrices:

$$\mathbf{Z_X} = \mathbf{AS} \quad \text{and} \quad \mathbf{Z_Y} = \mathbf{A\Theta S}. \tag{11}$$

In these expressions, $\mathbf{A}$ refers to the array response matrix, $\mathbf{S}$ refers to the source signal matrix, and $\boldsymbol{\Theta}$ refers to a diagonal matrix containing the shift-invariance term $\phi_i = e^{j2\pi\Delta\sin(\theta_i)}$. The two matrices $\mathbf{Z_X}$ and $\mathbf{Z_Y}$ are stacked vertically to give:

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z_X} \\ \mathbf{Z_Y} \end{bmatrix} = \mathbf{A}_Z\mathbf{S}, \quad \text{with} \quad \mathbf{A}_Z = \begin{bmatrix} \mathbf{A} \\ \mathbf{A\Theta} \end{bmatrix} \tag{12}$$

Then, the SVD of matrix $\mathbf{Z}$ is computed with the following:

$$\mathbf{Z} = \hat{\mathbf{U}}_z \boldsymbol{\Sigma}_z \hat{\mathbf{V}}_z^H, \tag{13}$$

due to the size of $\mathbf{U}_z$ which is $2M \times d$ it is concluded that there is a $d \times d$ invertible matrix $\mathbf{T}$, such that

$$\hat{\mathbf{U}}_z = \mathbf{A}_z\mathbf{T} = \begin{bmatrix} \mathbf{AT} \\ \mathbf{A\Theta T} \end{bmatrix}. \tag{14}$$

If $\hat{\mathbf{U}}_z$ is now splitted into into two $M \times d$ matrices in the same way that was done for $\mathbf{Z}$, the following is obtained:

$$\hat{\mathbf{U}}_z = \begin{bmatrix} \hat{\mathbf{U}}_x \\ \hat{\mathbf{U}}_y \end{bmatrix}. \tag{15}$$

Then, taking the pseudo-inverse of $\hat{\mathbf{U}}_x$ by using the expression in eq. (14) gives:

$$\hat{\mathbf{U}}_x^\dagger = (\mathbf{T}^H \mathbf{A}^H \mathbf{A} \mathbf{T})^{-1} \mathbf{T}^H \mathbf{A}^H = \mathbf{T}^{-1} \mathbf{A}^\dagger \tag{16}$$

Next, $\hat{\mathbf{U}}_x^\dagger$ from eq. (16) and $\hat{\mathbf{U}}_y$ from eq. (14) can be multiplied to have:

$$\hat{\mathbf{U}}_x^\dagger \hat{\mathbf{U}}_y = \mathbf{T}^{-1} \boldsymbol{\Theta} \mathbf{T}. \tag{17}$$

Expression in eq. (17) can be recognized as the eigendecomposition; therefore, from the eigenvalue matrix of $\boldsymbol{\Theta}$, first the ϕ_is then by using the expression $\phi_i = e^{j2\pi\Delta \sin(\theta_i)}$, the angles of θ_is can be estimated. This concludes the theory of ESPRIT algorithm for the estimation of directions.

To implement the ESPRIT algorithm in MATLAB, the function `esprit(X,d)` is implemented with the inputs `X`, `d` and the output `theta`. `X` is the the data matrix $\mathbf{X}$, `d` is the number of sources with $d = 2$ and `theta` is the $\hat{\boldsymbol{\theta}}$ which contains DOA estimates for both sources. The following Table 1 presents the DOA estimations for low and high SNR scenarios for the ESPRIT algorithm.

Table 1: True and estimated DOAs for Source 1 and 2 using ESPRIT at different SNR levels.

SNR (dB)	True DOAs (degrees)	Estimated DOAs (degrees)
20	-20.0000 30.0000	-20.1290 30.2355
100	-20.0000 30.0000	-20.0000 30.0000

From Table 1, it can be stated that the ESPRIT algorithm performs well in estimating the DOAs in both low and high SNR cases. At a lower SNR of 20 dB there are small estimation errors due to the presence of noise. This is because noise affects the accuracy of the estimation more significantly in low SNR cases than in higher SNR cases. However, at very high SNR (100 dB), the estimations are accurate and they reflect the true DOAs perfectly up to an order ambiguity.

2.3 Estimation of Frequencies

The ESPRIT algorithm can be used to estimate frequencies, as well. To do so, the special structure that is present in the $\mathbf{X}$ for the model given in eq. (8) is used. The size of $\mathbf{X}$ was $M \times N$; therefore, it can be said that the row space of $\mathbf{X}$ spans the different antennas and the column space of $\mathbf{X}$ spans the different time samples. For the DOA estimation, the main interest was the spatial arrangement of antennas; therefore, the structure of $\mathbf{X}$ was taken into account as it was. Now for the frequency estimation, the time structure of $\mathbf{X}$ has to be used, to correctly capture the frequency components. For this purpose the steps that were carried out in the ESPRIT for direction estimation has to be carried out for the frequency estimation as well but for the column space of $\mathbf{X}$. Taking the transpose of $\mathbf{X}$ in the beginning suffices so that time samples span the rows. The rest of the algorithm is the same but of course the mapping from the eigenvalue matrix of $\boldsymbol{\Theta}$ to the estimations are done differently. The angular frequencies in radians per sample are computed with $\omega_i = \text{angle}(\phi_i)$ and then the normalized frequencies are found with $f = |\omega/2\pi|$. This concludes the theory of ESPRIT algorithm for the estimation of frequencies.

To implement the modified ESPRIT algorithm for the estimation of frequency, the function of `espritfreq(X,d)` is implemented with the inputs `X`, `d` and the output `f`. The output is the $\hat{\mathbf{f}}$ which contains normalized frequency estimates for both sources. The following Table 2 presents the normalized frequency estimations for low and high SNR scenarios for the ESPRIT algorithm.

Table 2: True and estimated normalized frequencies for Source 1 and 2 using ESPRIT at different SNR levels.

SNR (dB)	True Frequencies	Estimated Frequencies
20	0.1000	0.0995
	0.3000	0.2998
100	0.1000	0.1000
	0.3000	0.3000

From Table 2, it can be stated that the ESPRIT algorithm modified to estimate the normalized frequencies performs well in both low and high SNR cases. The same analysis discussed in table 1 applies here as well. Overall, the results for both table 1 and table 2, show that when the signal power is high relative to the noise power, the estimations with ESPRIT are accurate, reflecting the true parameter values as the effect of noise is minimized.

2.4 Joint Estimation of Directions and Frequencies

Instead of estimating the angles and frequencies of d sources separately, as shown with ESPIRT, they can also be found simultaneously. The joint estimation uses both the spatial and temporal properties of the signals received by a ULA of antennas. By applying spatial smoothing, the original data is transformed in a way that reduces the correlation between signals and increases the rank as below:

$$\mathbf{X}_m = \begin{bmatrix} \mathbf{x}(0) & \mathbf{x}(1) & \cdots & \mathbf{x}(N-1) \\ \mathbf{x}(1) & \mathbf{x}(2) & \cdots & \mathbf{x}(N) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}(m-1) & \mathbf{x}(m) & \cdots & \mathbf{x}(N+m-2) \end{bmatrix}. \tag{18}$$

This matrix can be rewritten, using the array response matrix and the signal source vector used for generating the signal model. These definitions are shown below:

$$\begin{aligned} \mathbf{A}_\theta &= \begin{bmatrix} \mathbf{a}(\theta_1) & \mathbf{a}(\theta_2) & \cdots & \mathbf{a}(\theta_d) \end{bmatrix}, \\ \mathbf{\Phi} &= \mathrm{diag}(e^{j2\pi f_1}, e^{j2\pi f_2}, \ldots, e^{j2\pi f_d}), \\ \mathbf{s}(t) &= \begin{bmatrix} s_1(t) \\ s_2(t) \\ \vdots \\ s_d(t) \end{bmatrix}. \end{aligned} \tag{19}$$

Using these definitions, the smoothed data matrix $\mathbf{X}$ can be defined as shown in eq. (20) and further simplified to eq. (21):

$$\mathbf{X}_m = \begin{bmatrix} \mathbf{A}_\theta \mathbf{s}(0) & \mathbf{A}_\theta \mathbf{\Phi} \mathbf{s}(1) & \cdots \\ \mathbf{A}_\theta \mathbf{\Phi} \mathbf{s}(1) & \mathbf{A}_\theta \mathbf{\Phi}^2 \mathbf{s}(2) & \cdots \\ \vdots & \vdots & \ddots \\ \mathbf{A}_\theta \mathbf{\Phi}^{m-1} \mathbf{s}(m-1) & \mathbf{A}_\theta \mathbf{\Phi}^{m} \mathbf{s}(m) & \cdots \end{bmatrix}, \tag{20}$$

$$\mathbf{X}_m = \begin{bmatrix} \mathbf{A}_\theta \\ \mathbf{A}_\theta\mathbf{\Phi} \\ \vdots \\ \mathbf{A}_\theta\mathbf{\Phi}^{m-1} \end{bmatrix} \begin{bmatrix} \mathbf{s}(0) & \mathbf{\Phi}\mathbf{s}(1) & \cdots & \mathbf{\Phi}^{(N-1)}\mathbf{s}(N-1) \end{bmatrix} = (\mathbf{F}_\phi \circ \mathbf{A}_\theta)(\mathbf{F} \circ \mathbf{S}). \tag{21}$$

Now the data model $\mathbf{X}_m$ is written in a Khatri-Rao product structure. By performing the singular value decomposition on this smoothed matrix, $\mathbf{X}_m$ is decomposed into three submatrices,

$$\mathbf{X}_m = \mathbf{U}\Sigma\mathbf{V}^H. \tag{22}$$

The left singular vectors $\mathbf{U}$, capture the principal components of the smoothed data matrix. More importantly, as shown with ESPRIT, the first d columns of $\mathbf{U}$, corresponding to the largest singular values, span the signal subspace. Thus, the left singular vectors $\mathbf{U}_r$ represent the smoothed array response and are comparable to the first part of the simplified matrix representation (eq. (21)) as:

$$\mathbf{U}_r \approx \begin{bmatrix} \mathbf{A}_\theta \\ \mathbf{A}_\theta\mathbf{\Phi} \\ \vdots \\ \mathbf{A}_\theta\mathbf{\Phi}^{m-1} \end{bmatrix}. \tag{23}$$

The $\mathbf{U}_r$ can be split in two manners resulting in matrices which have similar properties as in the previous direction and frequency estimation with ESPRIT which is given below:

$$\begin{bmatrix} \mathbf{U}_{x\theta} \\ \mathbf{U}_{y\theta} \end{bmatrix} = \begin{bmatrix} \mathbf{R} \\ \mathbf{R}\Theta \end{bmatrix} \mathbf{T} \qquad \begin{bmatrix} \mathbf{U}_{x\phi} \\ \mathbf{U}_{y\phi} \end{bmatrix} = \begin{bmatrix} \mathbf{R} \\ \mathbf{R}\Phi \end{bmatrix} \mathbf{T}. \tag{24}$$

For angle estimation the matrix $\mathbf{U}_{x\theta}$ is formed by taking the first $M-1$ rows from each block of $\mathbf{U}_r$. $\mathbf{U}_{y\theta}$ is formed by taking the second to M^{th} rows from each block of $\mathbf{U}_r$. These are shown below:

$$\mathbf{U}_{x\theta} = \begin{bmatrix} \mathbf{U}_r(1:M-1,:) \\ \mathbf{U}_r(M+1:2M-1,:) \\ \vdots \\ \mathbf{U}_r((m-1)M+1:mM-1,:) \end{bmatrix}, \tag{25}$$

$$\mathbf{U}_{y\theta} = \begin{bmatrix} \mathbf{U}_r(2:M,:) \\ \mathbf{U}_r(M+2:2M,:) \\ \vdots \\ \mathbf{U}_r((m-1)M+2:mM,:) \end{bmatrix}. \tag{26}$$

For frequency estimation, the splitting will result in $\mathbf{U}_{x\phi}$ and $\mathbf{U}_{y\phi}$:

$$\mathbf{U}_{x\phi} = \mathbf{U}_r(1:(M*(m-1)),:), \tag{27}$$

$$\mathbf{U}_{y\phi} = \mathbf{U}_r((M+1):(M*m),:). \tag{28}$$

Similar to before, the splitted matrix vectors are projected onto each other to incorporate the difference in directions and frequencies:

$$\mathbf{X}_\theta = \mathbf{U}_{x\theta}^\dagger \mathbf{U}_{y\theta}, \tag{29}$$

$$\mathbf{X}_{\phi} = \mathbf{U}_{x\phi}^{\dagger}\mathbf{U}_{y\phi}. \tag{30}$$

Finally, joint diagonalization is performed to find the common matrix $\mathbf{T}$. This matrix $\mathbf{T}$ can then be used to simultaneously estimate both the directions and frequencies of the signals as below:

$$\mathbf{X}_{\theta} = \mathbf{T}^{-1}\Theta\mathbf{T}, \tag{31}$$

$$\mathbf{X}_{\phi} = \mathbf{T}^{-1}\Phi\mathbf{T}. \tag{32}$$

This concludes the theory of joint estimation algorithm.

To implement the joint estimation algorithm to jointly estimate the directions and frequencies, the function of `joint(X,d,m)` is implemented with the inputs `X`, `d`, and the smoothing factor of `m`. This function has two outputs `theta` for DOAs and `f` for normalized frequencies. The following Table 3 presents the parameter estimations for low and high SNR scenarios for the joint estimation algorithm.

Table 3: True and estimated DOAs and normalized frequencies for Source 1 and 2 using joint estimation algorithm at different SNR levels.

SNR (dB)	**Parameter**	**True Values**	**Estimated Values**
20	DOAs (degrees)	-20.0000 30.0000	-20.0389 29.9885
	Frequencies	0.1000 0.3000	0.1014 0.2987
100	DOAs (degrees)	-20.0000 30.0000	-20.0000 30.0000
	Frequencies	0.1000 0.3000	0.1000 0.3000

From Table 3 it is seen that the joint estimation of the parameters is successful. At a lower SNR of 20 dB there are small estimation errors for both of the parameters due to the presence of noise. On the other hand, at very high SNR (100 dB), the estimations are accurate and they match with the true parameter values.

2.5 Comparison

The estimation performance of the algorithms of ESPRIT for the DOA and ESPRIT for the normalized frequency and joint estimation for both DOA and normalized frequency can now be compared. The comparisons are carried out again for $d = 2$ sources but with the values $M = 3$, $N = 20$, $\boldsymbol{\theta} = [-20, 30]^T$ and $\mathbf{f} = [0.1, 0.12]^T$. The mean values and standard deviations of the estimated angles and frequencies are plotted as a function of SNR from 0 dB to 20 dB for $k = 1000$ test runs with independent complex Gaussian noise values. Figures 5 and 6 present the mean of the estimations over the trial runs for every SNR value between 0 and 20 dB, Figures 7 and 8 present the mean of the estimations over the trial runs for every SNR value between 0 and 20 dB, and Figures 9 and 10 present the mean values with their respective error bars for 4 consecutive SNR values within the same interval.

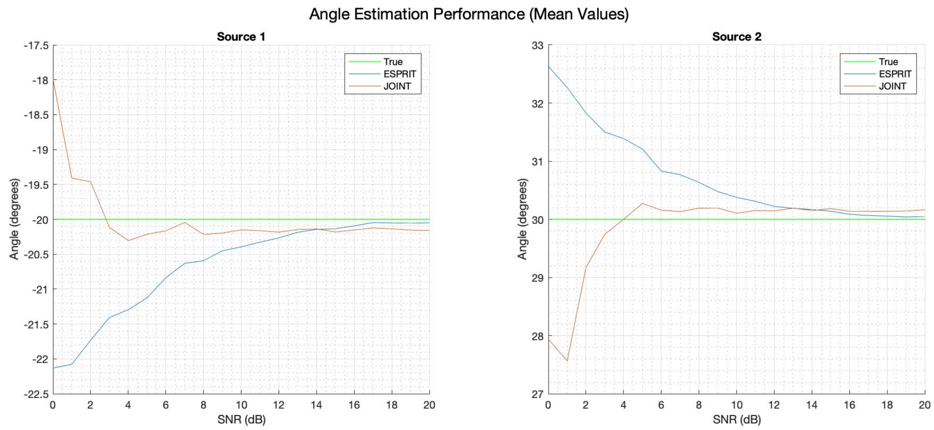

Figure 5: DOA estimation performance as a function of SNR with mean values over $k = 1000$ trial runs.

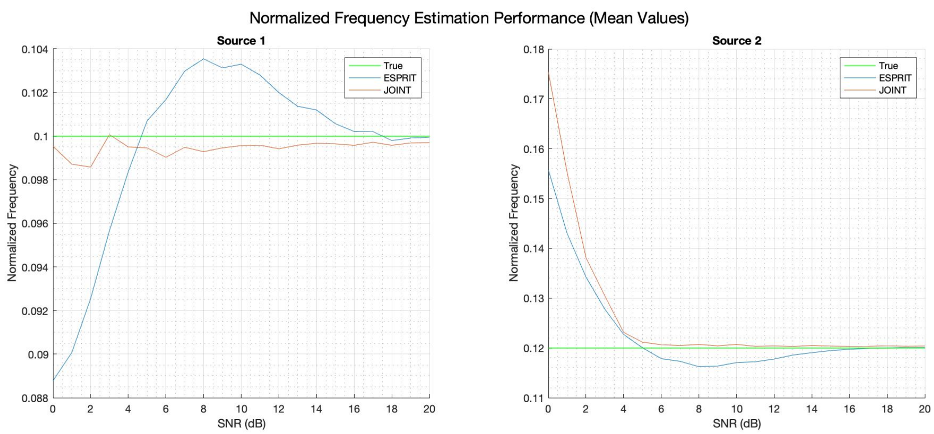

Figure 6: Normalized frequency estimation performance as a function of SNR with mean values over $k = 1000$ trial runs.

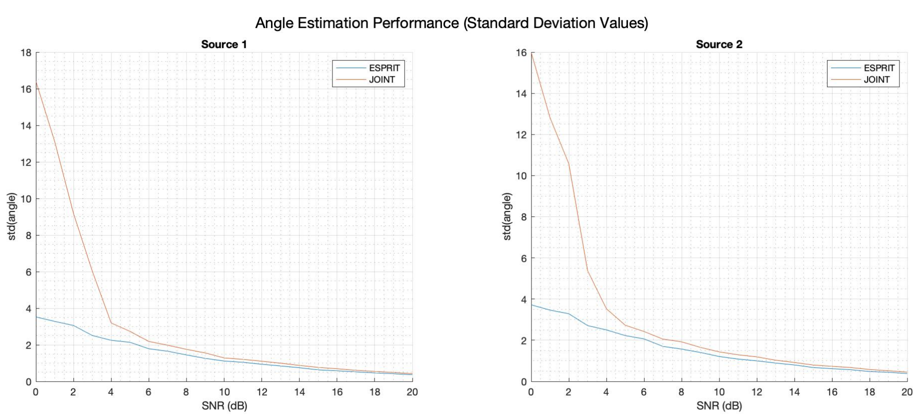

Figure 7: DOA estimation performance as a function of SNR with standard deviation values over $k = 1000$ trial runs.

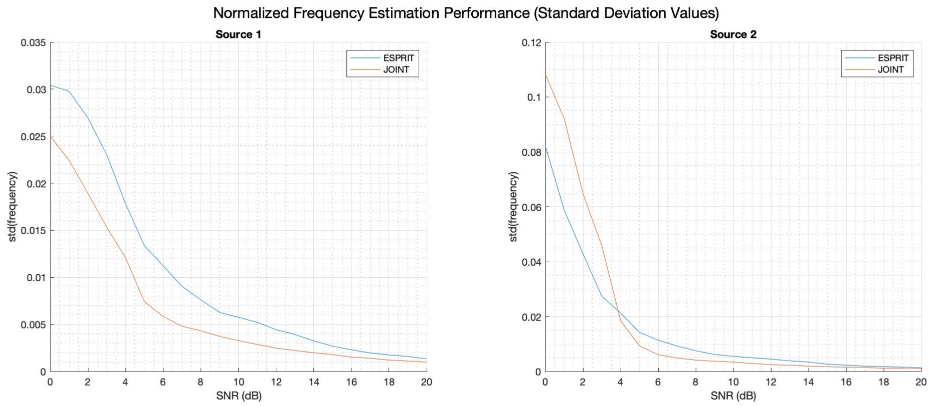

Figure 8: Normalized frequency estimation performance as a function of SNR with standard deviation values over $k = 1000$ trial runs.

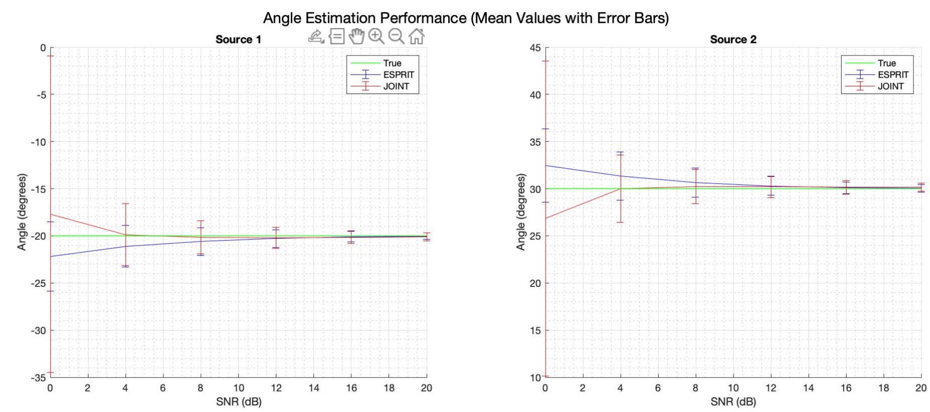

Figure 9: DOA estimation performance as a function of SNR with mean values and error bars over $k = 1000$ trial runs.

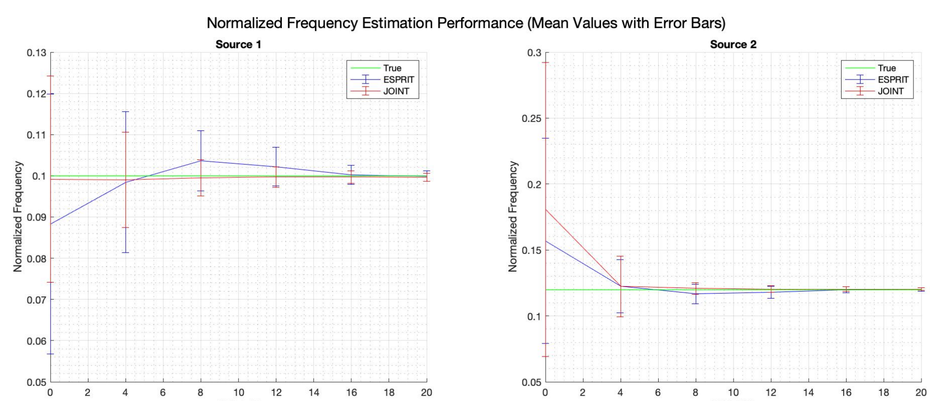

Figure 10: Normalized frequency estimation performance as a function of SNR with mean values and error bars over $k = 1000$ trial runs.

When these plots are analyzed there is a clear trend that shows better estimation results with an increase in SNR for both of the sources and for both of the algorithms ESPRIT and joint estimation algorithms. Additionally, the standard deviation curves in Figures 7 and 8 indicate that as the signal power increases relative to the noise, the variability in the estimations decreases. This results in more confident and reliable estimations. When the Figures 5, 7 are observed, although the joint estimation has an overall high standard deviation, when the means of the estimates are computed the estimates converge to the actual DOA values faster than the ESPRIT algorithm which suggest that there are some outliers in the estimates. When the Figures 6, 8 are observed, the results show a better convergence to the actual value of the normalized frequencies with the joint estimation as well. For the standard deviation curves, the performance varies between sources.

Two zero-forcing beamformers can be derived, one based on the estimates of the function `esprit(.)` and the other based on the estimates of the function `espritfreq(.)`. In the case of using the estimates of the angle of arrival, the steering vectors can be approximated. The weight matrix can be easily derived as $\mathbf{W}^H = \hat{\mathbf{A}}^\dagger$. Using the estimates of frequencies, the signal sources can be approximated. In combination with the measurement matrix X, the approximated array response matrix can be derived as $\hat{\mathbf{A}} = \mathbf{X}\hat{\mathbf{S}}^\dagger = \mathbf{X}\hat{\mathbf{S}}^H(\hat{\mathbf{S}}\hat{\mathbf{S}}^H)^{-1}$. To test the correctness of the two beamformers for two different SNRs, the norm differences between the source signal matrix $\mathbf{S}$ and $\hat{\mathbf{S}} = \mathbf{W}^H\mathbf{X}$. The following Table 4 presents these results.

Table 4: Norm differences for ESPRIT-based and ESPRITfreq-based recoveries at different SNR levels.

SNR (dB)	**ESPRIT-based Recovery**	**ESPRITfreq-based Recovery**
10	1.2540e+00	2.3977e+00
100	3.7382e-05	5.4573e-05

As expected, in case there is almost no noise (where SNR is 100 dB) $\mathbf{W}^H\mathbf{X}$ perfectly recovers the transmitted sources up to an order ambiguity. For the low SNR case with 10 dB, the norm difference is of course higher due to the effect of noise.

To understand how well the beamformer can focus on the desired signal direction and how it suppresses the interference from undesired directions spatial response can be viewed. Figure 11 below shows the spatial responses for both zero-forcing beamformers.

Figure 11: Spatial response of the zero-forcing beamformers.

As can be seen, both beamformers show high sensitivity for the directions of the sources. There is a better suppression of the directions between the -20 and 30 degrees for the frequency-based zero-forcing beamformer in comparison with the direction-based beamformer. However, there is a higher variability for generating the frequency-based weights, where the direction-based beamformer is constant when generating its weights. This behaviour indicates that the estimation of frequencies using ESPRIT freq is sensitive to noise, and can be explained by the relatively low SNR ratio. Small changes in the data can lead to bigger variations in the estimated frequencies. Direction estimation depends on the spatial correlation matrix, which is less noise-sensitive.

3 Channel Equalization

For the channel equalization part the modeling and reconstruction of transmitted data symbols over a coded channel is considered. This involves constructing a data matrix from sampled received signals and applying both zero-forcing and Wiener equalizers to analyze their performance. Throughout this part, the following setup is considered:

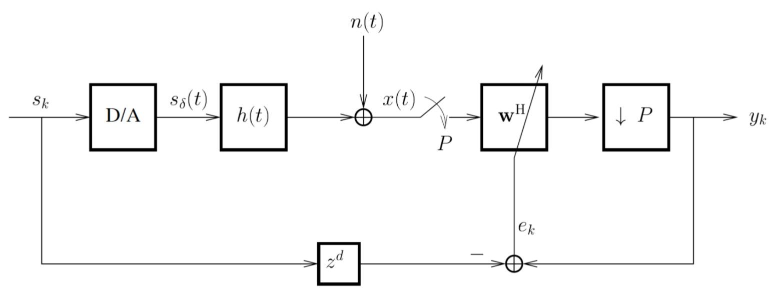

Figure 12: Channel equalization setup.

In Figure 12, data symbols $s_k = [s_0, s_1, \dots, s_{N-1}]^T$ are transmitted through a channel characterized by the impulse response $h(t)$ and are affected by additive noise $n(t)$. The impulse response of the channel is the following:

$$h(t) = \begin{cases} 1 & \text{for } t \in [0, 0.25) \\ -1 & \text{for } t \in [0.25, 0.5) \\ 1 & \text{for } t \in [0.5, 0.75) \\ -1 & \text{for } t \in [0.75, 1) \end{cases}. \tag{33}$$

Furthermore, the channel is QPSK-modulated with $s_k \in \{\pm 1/\sqrt{2} \pm j/\sqrt{2}\}$ and the additive white noise $n(t)$ is modelled to be complex zero-mean Gaussian noise with standard deviation σ. With all these the received signal $x(t)$ can be written as:

$$x(t) = \sum_k h(t-k)s_k + n(t). \tag{34}$$

In the following subsections the signal model structure and the application of zero-forcing and Wiener equalizers are explored.

3.1 Signal Model

The goal of this part of the assignment is to reconstruct the transmitted data symbols s_k. This will be done by first implementing a function to generate $\mathbf{x} = [x(0)\ x(1/P)\ \cdots\ x((N-1)/P)]^T$ which is obtained by sampling $x(t)$ at rate $1/P$. P is called the oversampling factor. The oversampling will give more detailed information about the signal and make it possible to capture the effects of the channel more accurately. The function `gendata_conv(s, P, N, sigma)` delivers the output `x`.

Later, a matrix $\mathbb{X}$ where each row represents the received signal at different time shifts is constructed. Zero-forcing and Wiener equalizers will use this structure to better separate the distortion caused by propagating through the channel from the transmitted symbols. This matrix and its structure can be seen below:

$$\mathbb{X} = \begin{bmatrix} x(0) & x(1) & \cdots & x(N-2) \\ x\left(\frac{1}{P}\right) & x\left(1+\frac{1}{P}\right) & \cdots & x\left(N-2+\frac{1}{P}\right) \\ \vdots & \vdots & \ddots & \vdots \\ x\left(1+\frac{P-1}{P}\right) & x\left(2+\frac{P-1}{P}\right) & \cdots & x\left(N-1+\frac{P-1}{P}\right) \end{bmatrix}. \tag{35}$$

Below, the rank of $\mathbb{X}$ dependent on the σ and P values can be viewed in Table 5.

Table 5: Rank of $\mathbb{X}$ and its dependence on σ and P.

σ	P	**Rank**
0	4	2
	8	2
0.5	4	8
	8	16

Since $\mathbb{X}$ has $2P$ rows, the maximum possible rank is $2P$. This is seen in the bottom row of Table 5. The rows will be linearly independent when there is noise present making every row unique. In the case of no noise, the received signal $x(t)$ is a deterministic function of the transmitted symbols s_k and the channel impulse response $h(t)$. Therefore, the time-shifted versions, due to oversampling, are not linearly independent but are based on the same underlying signal. This leads to a matrix which contains linear dependencies among the rows of $\mathbb{X}$, and therefore results in a rank of 2.

3.2 Zero-Forcing and Wiener Equalizer

In this part the zero-forcing and Wiener receivers for the data matrix $\mathbb{X}$ defined in eq. (35) is computed assuming the channel impulse response $h(t)$ and the standard deviation of the noise σ are known. The relationship between the received signal matrix $\mathbb{X}$, the channel matrix $\mathbf{H}$, and the symbol matrix $\mathbf{S}$ is:

$$\mathbb{X} = \mathbf{HS} \tag{36}$$

The Zero-Forcing equalizer tries to invert the channel effects by applying the pseudoinverse of the channel matrix $\mathbf{H}$ and by this minimizing $\arg\min_{\hat{\mathbf{s}}_{ZF}} \|\mathbf{X} - \mathbf{AS}\|_F^2$:

$$\hat{\mathbf{s}}_{ZF} = \mathbf{H}^{\dagger}\mathbb{X}. \tag{37}$$

The Wiener equalizer minimizes the mean square error between the estimated and actual transmitted symbols by using the autocorrelation matrix of the channel matrix $\mathbf{H}$:

$$\mathbf{R}_{hh} = \frac{1}{N}\mathbf{HH}^H \tag{38}$$

The optimal Wiener filter weights are computed as:

$$\mathbf{w}_{opt} = (\mathbf{R}_{hh} + \sigma^2\mathbf{I})^{-1}\mathbf{H} \tag{39}$$

The estimated symbols using the Wiener equalizer are obtained by:

$$\hat{\mathbf{s}}_{Wiener} = \mathbf{w}_{opt}^H\mathbf{X} \tag{40}$$

Figures 13 and 14 below are plots that show the estimated symbols for the zero-forcing and Wiener equalizers for an oversampling factor of $P = 4$ and $P = 8$.

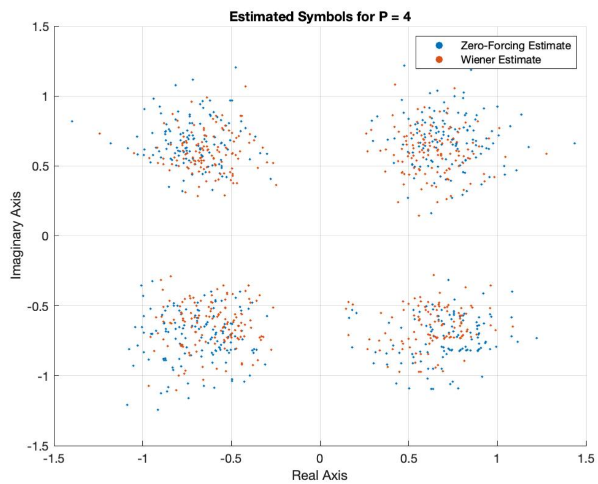

Figure 13: Estimated symbols for $P = 4$.

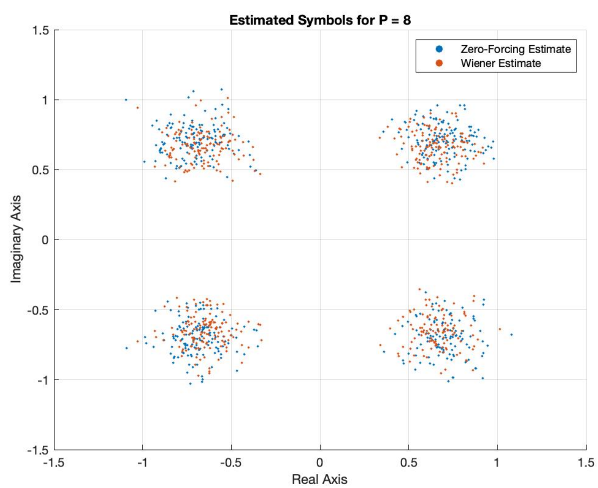

Figure 14: Estimated symbols for $P = 8$.

From Figures 13 and 14 it can be seen that the differences between these two equalizers are very small. Their clustering is very similar and seems shifted. It could be argued that the Wiener equalizer is clustered a bit better due to its fewer outliers. When P is doubled it creates a really big change in the estimates. Higher P gives rise to a higher performance for both equalizers due to their tighter clustering. This is expected because a higher oversampling reduces the impact of the noise, and therefore gives a better representation of the signal, and thus a tighter clustering.

When the delay of the data symbol sequence $\mathbf{s}$ is considered, focusing the receivers on a particular shift of $\mathbf{s}$ does not really change any results for the clustering of the equalizers since the QPSK symbols are generated in a random fashion anyways. However, if the delay refers to the received signals $\mathbf{x}$, then the interpretation of a good delay is a bit different. If the delay is implemented into the system by delaying the signal $\mathbf{x}$ and then zero-padding it to make it fit with the desired length, the delay should at least be in the multiples of the sampling factor P. This ensures the structure in $\mathbb{X}$ to be preserved leading to a good reconstruction of the data symbols. In such a case, there exists 5 clusters instead of 4, because of the zero-padding. Therefore, the optimal delay in this problem is having no delay.

4 Conclusion

This report comprehensively explored the estimation of directions and frequencies of sinusoidal signals coming into a ULA structure with multiple antennas and it also examined channel equalization techniques. The effect of setup parameters were observed by looking at the singular values of the received signal matrix. For the estimation of the DOA and normalized frequency, ESPRIT algorithm in two modified forms and a joint angle and frequency estimation algorithm that makes use of the data model's Khatri-Rao structure was employed. The comparison of these algorithms were carried out. In the context of channel equalization, both zero-forcing and Wiener equalizers were applied to reconstruct transmitted data symbols over a QPSK-modulated channel. The Wiener equalizer demonstrated slightly better performance in clustering of the estimated symbols. Also, it was shown that oversampling improved the performance of both equalizers. Overall, the report demonstrated the practical application of theoretical array processing concepts.